Legend

F = page fault
B = process blocked
R = process ready
Pa(Xb) = processed page b of process a (X = R read, W write)
f= frame number

Clock - Fixed-Local Replacement, 30 frames (7 per process, 2 not allocated), quantum 3, input: Process1.txt ... Process4.txt

WB = dirty page(s) written back to the disk (1 time unit): a dirty victim page (paid by the faulting process) or, after the last execution of a process, all its dirty pages in parallel
Orange blocked-indicator cell = the write-back time unit (the process finishes at the end of the orange cell if it is the last write-back)
* = dirty page (modified by a write after it was loaded)
Orange frame cell = frame locked while its dirty page is written back (also all dirty frames of a terminating process)

Ready Queue

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
time																																				
CPU events	P1F				P1(W1)	P1F					P1(R2)	P1F						P1(W3)	P1F					P1(R4)	P1F				P1(R5)							
	P2F					P2(W5)	P2F								P2(R4)	P2F				P2(R3)	P2F				P2(W2)	P2F							P2(R1)			
	P3F						P3(R1)	P3(R1)	P3(R1)			P3(R1)	P3(R1)	P3(R1)			P3(R1)																			
	P4F									P4(R1)	P4F					P4(W2)	P4F				P4(R3)	P4(R2)	P4(W1)		P4F				P4(R4)	P4(R4)	P4(W4)					
Process status events	P1B				P1R	P1B				P1R		P1B				P1R			P1B				P1R		P1B				P1R	P1WB						
	P2B				P2R		P2B				P2R					P2B				P2R	P2B				P2R	P2B				P2R				P2WB		
	P3B				P3R																															
	P4B				P4R						P4B				P4R		P4B				P4R				P4B				P4R				P4WB			
Blocked Indicator P1																																				
Blocked Indicator P2																																				
Blocked Indicator P3																																				
Blocked Indicator P4																																				

Allocated to P1

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
f1					1	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*						
f2										2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2						
f3																3	3	3	3*	3*	3*	3*	3*	3*	3*	3*	3*	3*	3*	3*						
f4																							4	4	4	4	4	4	4	4						
f5																													5	5						
f6																																				
f7																																				

Allocated to P2

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
f8					5	5	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*	5*		
f9											4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4		
f10																				3	3	3	3	3	3	3	3	3	3	3	3	3	3	3		
f11																									2	2*	2*	2*	2*	2*	2*	2*	2*	2*		
f12																														1	1	1	1	1		
f13																																				
f14																																				

Allocated to P3

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
f15					1	1	1	1	1	1	1	1	1	1	1	1	1																			
f16																																				
f17																																				
f18																																				
f19																																				
f20																																				
f21																																				

Allocated to P4

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
f22					1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1*	1*	1*	1*	1*	1*	1*	1*	1*	1*			
f23															2	2	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*	2*			
f24																			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3			
f25																													4	4	4	4	4*			
f26																																				
f27																																				
f28																																				

Not Allocated

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
f29																																				
f30																																				

Summary

PID	Turnaround time	Faults	Fault times	Write-backs
P1	30	5	0, 5, 11, 18, 24	1
P2	34	5	0, 6, 15, 20, 25	1
P3	17	1	0	0
P4	33	4	0, 10, 16, 24	1
	avr:	28.5		